STATISTICS IS THE GRAMMAR OF SCIENCE

PROBABILITY AND STATISTICS

LECTURE – 27

REGRESSION ANALYSIS
MULTIPLE LINEAR REGRESSION

PREPARED BY
HAZBER SAMSON
FAST-NUCES ISLAMABAD

REGRESSION ANALYSIS

INTRODUCION: At the beginning of the nineteenth century, **Legendre and Gauss** published papers on the *method of least squares*, which implemented the earliest form of what is now known as *linear regression*. The term *regression* was first introduced by **Francis Galton**. In a famous paper, Galton found that, although there was a tendency for tall parents to have tall children and for short parents to have short children, the average height of children born of parents of a given height tended to move or "regress" toward the average height in the population as a whole.1 In other words, the height of the children of unusually tall or unusually short parents tends to move toward the average height of the population. Galton's *law of universal regression* was confirmed by his friend Karl Pearson, who collected more than a thousand records of heights of members of family groups.2 He found that the average height of sons of a group of tall fathers was less than their fathers' height and the average height of sons of a group of short fathers was greater than their fathers' height, thus "regressing" tall and short sons alike toward the average height of all men. In the words of Galton, this was "regression to mediocrity. The modern interpretations of regression is, as follows however,

DEFINITION: Regression analysis is a statistical technique for investigating and modeling the relationship between variables.

Applications of regression are numerous and occur in almost every field, including engineering, the physical and chemical sciences, economics, management, life and biological sciences, and the social sciences. In fact, regression analysis may be the most widely used statistical technique.

Usually, Researchers want more than just finding the best linear approximation of one variable given a set of others. They want valid statistical relationships. They want to draw conclusions about what happens if one of the variables actually changes. That is: they want to say something about things that are not observed (yet). In this case, they want to find a fundamental relationship. To do this it is assumed that there is a general relationship that is valid for all possible observations from a well-defined population. Restricting attention to linear relationships, we specify a statistical model as linear regression model. Some examples of analyses using regression models include the following:

• Estimating weight gain by the addition to children's diet of different amounts of various dietary supplements
• Predicting scholastic success (grade point ratio) based on students' scores on an aptitude or entrance test
• Estimating amounts of sales associated with levels of expenditures for various types of advertising
• Predicting fuel consumption for home heating based on daily temperatures and other weather factors
• Estimating changes in interest rates associated with the amount of deficit spending

Regression analysis is used to predict the value of one variable on the basis of the other variables. Regression analysis is generally classified into two kinds, **simple** and **multiple**. The regression will be **simple** if it involves only one independent variable and it will be **multiple** if it involves more than one independent variables.

MULTIPLE LINEAR REGRESSION MODEL (MLRM)

MULTIPLE LINEAR REGRESSION MODEL USING MATRICES APPROACH

A mathematical expression that shows the dependence of one variable upon more than one independent variables is known as Multiple Linear Regression Model.

The standard form of multiple linear regression Model can be written as

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + ... + \beta_k X_k + \varepsilon$$

In general if we have n observations for the above MLRM then it can be written as

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + ... + \beta_k X_{ik} + \varepsilon_i$$

Here X_{ij} represents i^{th} observation of the jth independent variable. Where $i = 1,2,...,n$ *and* $j = 1,2,...,k$

If we expand the above Model it becomes

$$Y_1 = \beta_0 + \beta_1 X_{11} + \beta_2 X_{12} + ... + \beta_k X_{1k} + \varepsilon_1$$
$$Y_2 = \beta_0 + \beta_1 X_{21} + \beta_2 X_{22} + ... + \beta_k X_{2k} + \varepsilon_2$$
$$. \quad . \quad . \qquad . \quad . \quad .$$
$$. \quad . \quad . \qquad . \quad . \quad .$$
$$Y_n = \beta_0 + \beta_1 X_{n1} + \beta_2 X_{n2} + ... + \beta_k X_{nk} + \varepsilon_n$$

In matrix notation

$$\begin{pmatrix} y_1 \\ y_2 \\ . \\ . \\ . \\ y_n \end{pmatrix} = \begin{pmatrix} 1 & x_{11} & x_{12} & . & . & x_{1k} \\ 1 & x_{21} & x_{22} & . & . & x_{2k} \\ . & . & . & . & . & . \\ . & . & . & . & . & . \\ . & . & . & . & . & . \\ 1 & x_{n1} & x_{n2} & . & . & x_{nk} \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ . \\ . \\ . \\ \beta_k \end{pmatrix} + \begin{pmatrix} e_1 \\ e_2 \\ . \\ . \\ . \\ e_n \end{pmatrix}$$

$$\Rightarrow Y = XB + E$$

where

$$Y = \begin{pmatrix} y_1 \\ y_2 \\ . \\ . \\ y_n \end{pmatrix}_{n\times 1}, \quad X = \begin{pmatrix} 1 & x_{11} & x_{12} & . & . & x_{1k} \\ 1 & x_{21} & x_{22} & . & . & x_{2k} \\ . & . & . & . & . & . \\ . & . & . & . & . & . \\ 1 & x_{n1} & x_{n2} & . & . & x_{nk} \end{pmatrix}_{n\times(k+1)}, \quad B = \begin{pmatrix} \beta_0 \\ \beta_1 \\ . \\ . \\ \beta \end{pmatrix}_{(k+1)\times 1} \quad and \quad E = \begin{pmatrix} e_1 \\ e_2 \\ . \\ . \\ e_n \end{pmatrix}_{n\times 1}$$

Note that
Y is a vector of observations on the dependent variable *Y*
X is called **Data Matrix**
B is called **vector of parameters**
E is called **vector of error terms**

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ASSUMPTIONS OF MULTIPLE LINEAR REGRESSION MODEL

Consider the classical linear regression model (CLRM)

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + ... + \beta_k X_{ik} + \varepsilon_i$$

ASSUMPTION-1 LINEARITY *The regression model is linear in the parameters,* though it may or may not be linear in the variables. It means that the dependent variable can be calculated as a linear function of a specific set of independent variables, plus a disturbance term.

ASSUMPTION-2 NUMBER OF OBSERVATIONS The number of observations *n* must be greater than the number of parameters to be estimated: Alternatively, the number of observations must be greater than the number of explanatory variables.

ASSUMPTION-3 INDEPENDENT VARIABLES HAVE SOME VARIATION
By this assumption we mean that not all observations of any independent variable X are the same; at least one has to be different so that the sample Var(X) is not 0. Furthermore, there can be no outliers in the values of the *X* variable.

ASSUMPTION-4 EXPECTED VALUE OF THE ERROR TERM IS ZERO $E(\varepsilon_i) = 0 \ \ for\, all\, i.$
Mean value of error terms is zero. ie This means that the disturbance is a genuine disturbance, so if we took a large number of samples the mean disturbance will be zero.

ASSUMPTION-5 HOMOSCEDASTICITY OF ERROR TERMS $Var(\varepsilon_i) = \sigma^2 \ \ for\, all\, i.$
Variance of error terms is same for all observations. This means that all error terms have the same variance regardless of the value of X.

ASSUMPTION-6 INDEPENDENCE OF ERROR TERMS $Cov(\varepsilon_i, \varepsilon_j) = 0 \ \ for\, all\, i \neq j$
Error terms are independent of each other. So there is no correlation between two error terms. In short, the observations are sampled independently.

ASSUMPTION-7 FIXED VALUES OF INDEPENDENT VARIABLES $Cov\,(X_i, \varepsilon_i) = 0 \ \ for\, all\, i\,.$
Values taken by any regressor *X* may be considered fixed in repeated samples also X variable(s) and the error terms are independent of each other.

ASSUMPTION-8 NO EXACT LINEAR RELATIONSHIP BETWEEN REGRESSORS
There are no exact linear relationships among the sample values of any two or more of the explanatory variables.

ASSUMPTION-9 NORMALITY OF ERROR TERMS $\varepsilon_i \sim N(0, \sigma^2)\ \forall\ i$
Error terms are normally distributed with mean zero and variance σ^2

ASSUMPTION-10 NO SPECIFICATION BIAS
The model is correctly specified.

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FORMULAS FOR MLRM MATRICES APPROACH

For MLRM $Y = XB + E$

$1 - X^t X$

$$X^t X = \begin{bmatrix} n & \sum x_1 & \sum x_2 & . & . & \sum x_k \\ \sum x_1 & \sum x_1^2 & \sum x_1 x_2 & . & . & \sum x_1 x_k \\ \sum x_2 & \sum x_2 x_1 & \sum x_2^2 & . & . & \sum x_2 x_k \\ . & . & . & . & . & . \\ . & . & . & . & . & . \\ \sum x_k & \sum x_k x_1 & \sum x_k x_2 & . & . & \sum x_k^2 \end{bmatrix}$$

$2 -$ *Estimation of Parameters*

$$\hat{B} = (X^t X)^{-1}(X^t Y)$$

$3 -$ *Fitted Values*

$$\hat{y} = X\hat{\beta}$$

$4 -$ *Error Terms*

$$e = y - \hat{y}$$

$5 -$ *Sum of Squares of Errors*

$$SSE = e^t e$$

$6 -$ *Sum of Squares of* Regression

$$SSR = \hat{B}^t X^t Y - n\bar{Y}^2$$

$7 -$ *Sum of Squares Total*

$$SST = Y^t Y - n\bar{Y}^2$$

$8 -$ *Coefficient of Determination*

$$R^2 = \frac{SSR}{SST} \quad or \quad R^2 = 1 - \frac{SSE}{SST}$$

$9 -$ *Variance – Covariance Matrix*

$$C = \sigma_e^2 (X^t X)^{-1} \quad where \ \sigma_e^2 = \frac{SSE}{n - p}$$

$10 -$ *Correlation Matrix*

$$C.M = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} = \begin{bmatrix} 1 & r_{12} & r_{13} \\ r_{21} & 1 & r_{23} \\ r_{31} & r_{32} & 1 \end{bmatrix}$$

Note that $r_{11} = r_{22} = r_{33} = 1$ *and* $r_{12} = r_{21}$, $r_{13} = r_{31}$, $r_{23} = r_{32}$

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EXAMPLE The following data show the number of bedrooms, the number of baths, and the prices at which a random sample of eight one-family houses sold in a certain large housing development:

Number of bedrooms x_1	*Number of baths* x_2	*Price (dollars)* y
3	2	292,000
2	1	264,600
4	3	317,500
2	1	265,500
3	2	302,000
2	2	275,500
5	3	333,000
4	2	307,500

(a) Use the method of least squares to fit a linear equation that will enable us to predict the average sales price of a one-family house in the given housing development in terms of the number of bedrooms and the number of baths. Also Interpret the Results.
(b) Predict the sales price of a three-bedroom house with two baths in the subject housing development.
(c) Find Coefficient of Determination and Interpret it.

SOLUTION (a) We have multiple linear regression model given by

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon \quad or \quad Y = XB + E \qquad and \qquad \hat{B} = (X^t X)^{-1}(X^t y)$$

Here

$$y = \begin{pmatrix} 292000 \\ 264600 \\ 317500 \\ 265500 \\ 302000 \\ 275500 \\ 333000 \\ 307500 \end{pmatrix}, \quad X = \begin{pmatrix} 1 & 3 & 2 \\ 1 & 2 & 1 \\ 1 & 4 & 3 \\ 1 & 2 & 1 \\ 1 & 3 & 2 \\ 1 & 2 & 2 \\ 1 & 5 & 3 \\ 1 & 4 & 2 \end{pmatrix}, \quad X^t X = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 3 & 2 & 4 & 2 & 3 & 2 & 5 & 4 \\ 2 & 1 & 3 & 1 & 2 & 2 & 3 & 2 \end{pmatrix} \begin{pmatrix} 1 & 3 & 2 \\ 1 & 2 & 1 \\ 1 & 4 & 3 \\ 1 & 2 & 1 \\ 1 & 3 & 2 \\ 1 & 2 & 2 \\ 1 & 5 & 3 \\ 1 & 4 & 2 \end{pmatrix} = \begin{pmatrix} 8 & 25 & 16 \\ 25 & 87 & 55 \\ 16 & 55 & 36 \end{pmatrix}, \quad X^t y = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 3 & 2 & 4 & 2 & 3 & 2 & 5 & 4 \\ 2 & 1 & 3 & 1 & 2 & 2 & 3 & 2 \end{pmatrix} \begin{pmatrix} 292000 \\ 264600 \\ 317500 \\ 265500 \\ 302000 \\ 275500 \\ 333000 \\ 307500 \end{pmatrix} = \begin{pmatrix} 2357600 \\ 7558200 \\ 4835600 \end{pmatrix}$$

$$So \quad \hat{B} = (X^t X)^{-1}(X^t y) = \begin{pmatrix} 8 & 25 & 16 \\ 25 & 87 & 55 \\ 16 & 55 & 36 \end{pmatrix}^{-1} \begin{pmatrix} 2357600 \\ 7558200 \\ 4835600 \end{pmatrix} = \begin{pmatrix} 1.2738 & -0.238 & -0.202 \\ -0.238 & 0.3809 & -0.476 \\ -0.202 & -0.476 & 0.8452 \end{pmatrix} \begin{pmatrix} 2357600 \\ 7558200 \\ 4835600 \end{pmatrix} = \begin{pmatrix} 224928.57 \\ 15314.28 \\ 10957.14 \end{pmatrix}$$

So the fitted regression model is given by $\hat{Y} = 224929 + 15314X_1 + 10957X_2$

INTERPRETATION OF RESULTS OF THE MODEL

Interpretation of $\hat{\beta}_0 = 224929$**:** If a house has no bedrooms and no baths, even then its price will be $\$224929$.

Interpretation of $\hat{\beta}_1 = 15314$: If number of bedrooms in a house is increased by one then its price will be increased by $\$15314$ on average in the long run, keeping other factors as constant.

Interpretation of $\hat{\beta}_2 = 10957$: If number of baths in a house is increased by one then its price will be increased by $\$10957$ on average in the long run, keeping other factors as constant.

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(b) Prediction

We have to predict the sales price of a three-bedroom house with two baths in the subject housing development. i.e. $\hat{Y} = ?,\ X_1 = 3 \ \ and \ \ X_2 = 2$

Put the values in equation (B) we get

$$\hat{Y} = 224929 + 15314(3) + 10957(2) = \$292785$$

(c) Coefficient of Determination

Coefficient of Determination is given by

$$R^2 = \frac{SSR}{SST} \quad where \ \ SSR = \hat{B}^t X^t Y - n\bar{Y}^2 \quad and \quad SST = Y^t Y - n\bar{Y}^2$$

$$Here \ \ \bar{Y} = \frac{\sum Y}{n} = \frac{2357600}{8} = 294700$$

$$SSR == \hat{B}^t X^t Y - n\bar{Y}^2 = \begin{pmatrix} 224928.57 & 15314.29 & 10957.14 \end{pmatrix} \begin{pmatrix} 2357600 \\ 7558200 \\ 4835600 \end{pmatrix} - 8 \times (294700)^2 = 4239674286$$

$$Y^t Y = \begin{pmatrix} 292000 & 264600 & 317500 & 265500 & 302000 & 275500 & 333000 & 307500 \end{pmatrix} \begin{pmatrix} 292000 \\ 264600 \\ 317500 \\ 265500 \\ 302000 \\ 275500 \\ 333000 \\ 307500 \end{pmatrix} = 699123160000$$

$$\text{SST} = Y^t Y - n\bar{Y}^2 = 699123160000 - 8 \times (294700)^2 = 4338440000$$

$$\Rightarrow R^2 = \frac{SSR}{SST} = \frac{4239674286}{4338440000} = 0.9772347 = 0.98$$

Interpretation: $R^2 = 0.98$ shows that 98% variation in Price of a house is explained due to number of bedrooms and number of baths while remaining 2% is due to other factors.

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TESTING OF PARAMETERS OF MLRM

$$If \quad H_0 : \beta_j = 0 \quad vs \quad H_1 : \beta_j \neq 0$$

- we use **test statistic**

$$t = \frac{\hat{\beta}_j - \beta_j}{se\left(\hat{\beta}_j\right)} \qquad with \quad d.f = n - p$$

- we use **P-Value**

$$P = 2[\Phi(|t_0|)]$$

- The $(1-\alpha)$ 100% **confidence interval** for $\hat{\beta}_j$ is

$$\hat{\beta}_j - t_{\frac{\alpha}{2}(n-p)} \times se\left(\hat{\beta}_j\right) \leq \beta_j \leq \hat{\beta}_j + t_{\frac{\alpha}{2}(n-p)} \times se\left(\hat{\beta}_j\right)$$

NOTE THAT For $se\left(\hat{\beta}_j\right)$ we shall first calculate variance-covariance matrix, after finding variance-covariance matrix we take square root of the diagonal entries of the variance-covariance matrix and results are $se\left(\hat{\beta}_j\right)$.

Variance-Covariance matrix is given by

$C = \sigma_e^2 (X^t X)^{-1}$ *where* $\sigma_e^2 = \frac{SSE}{n-p}$

$\sigma_e^2 = \frac{SSE}{n-p}$

$SSE = e^t e$

$e = y - \hat{y}$

$\hat{y} = X\hat{B}$

1. $SSR = Sum\ of\ Squares\ \text{Re}gression = \hat{B}^t X^t Y - n\overline{Y}^2$
2. $SSE = Sum\ of\ Squares\ Errors = e^t e$
3. $SST = Sum\ of\ Squares\ Total = Y^t Y - n\overline{Y}^2$

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(a) Use the method of least squares to fit a linear equation that will enable us to predict the average sales price of a one-family house in the given housing development in terms of the number of bedrooms and the number of baths. Also Interpret the Results.
(b) Test Parameters of the Model using $\alpha = 0.05$.

SOLUTION (a) We have multiple linear regression model given by

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon \quad or \quad Y = XB + E \qquad and \qquad \hat{B} = (X^t X)^{-1}(X^t y)$$

Here

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So the fitted regression model is given by $\hat{Y} = 224929 + 15314X_1 + 10957X_2$

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Interpretation of $\hat{\beta}_2 = 10957$: If number of baths in a house is increased by one then its price will be increased by $\$10957$ on average in the long run, keeping other factors as constant.

(b) Testing of Parameters of the Model

First of all we shall find variance-covariance matrix, it is given by

$$C = \sigma_e^2 (X^t X)^{-1} \quad where \ \sigma_e^2 = \frac{SSE}{n-p}$$

$$Now \ \sigma_e^2 = \frac{SSE}{n-p}$$

$$SSE = e^t e$$

$$e = y - \hat{y}$$

$$Now \ \hat{y} = X\hat{B} = \begin{pmatrix} 1 & 3 & 2 \\ 1 & 2 & 1 \\ 1 & 4 & 3 \\ 1 & 2 & 1 \\ 1 & 3 & 2 \\ 1 & 2 & 2 \\ 1 & 5 & 3 \\ 1 & 4 & 2 \end{pmatrix} \begin{pmatrix} 224928.57 \\ 15314.28 \\ 10957.14 \end{pmatrix} = \begin{pmatrix} 292785.7 \\ 266514.3 \\ 319057.1 \\ 266514.3 \\ 292785.7 \\ 277471.4 \\ 334371.4 \\ 308100.0 \end{pmatrix}$$

$$e = y - \hat{y} = \begin{pmatrix} 292000 \\ 264600 \\ 317500 \\ 265500 \\ 302000 \\ 275500 \\ 333000 \\ 307500 \end{pmatrix} - \begin{pmatrix} 292785.7 \\ 266514.3 \\ 319057.1 \\ 266514.3 \\ 292785.7 \\ 277471.4 \\ 334371.4 \\ 308100.0 \end{pmatrix} = \begin{pmatrix} -785.71 \\ -1914.28 \\ -1557.14 \\ -1014.28 \\ 9214.28 \\ -1971.43 \\ -1371.43 \\ -600.00 \end{pmatrix}$$

$$SSE = e^t e = \begin{pmatrix} -785.71 & -1914.28 & -1557.14 & -1014.28 & 9214.28 & -1371.43 & 1371.43 & -600.00 \end{pmatrix} \begin{pmatrix} -785.71 \\ -1914.28 \\ -1557.14 \\ -1014.28 \\ 9214.28 \\ -1971.43 \\ -1371.43 \\ -600.00 \end{pmatrix}$$

$$SSE = e^t e = 98765714$$

$$\sigma_e^2 = \frac{SSE}{n-p} = \frac{98765714}{8-3} = \frac{98765714}{5} = 19753143$$

$$C = \sigma_e^2 (X^t X)^{-1} = 19753143 \begin{pmatrix} 1.2738 & -0.238 & -0.202 \\ -0.238 & 0.3809 & -0.476 \\ -0.202 & -0.476 & 0.8452 \end{pmatrix} = \begin{pmatrix} 1.2738 & -0.238 & -0.202 \\ -0.238 & 0.3809 & -0.476 \\ -0.202 & -0.476 & 0.8452 \end{pmatrix} = \begin{pmatrix} 25161741 & -4703129 & -3997660 \\ 4703129 & 7525007 & -9406259 \\ -3997660 & -9406259 & 16696109 \end{pmatrix}$$

So $\text{var}(\hat{\beta}_0) = 25161741, \quad \text{var}(\hat{\beta}_1) = 7525007, \quad \text{var}(\hat{\beta}_2) = 16696109$

Taking square root of the var*iances we get*

$$se(\hat{\beta}_0) = 5016.15, \qquad se(\hat{\beta}_1) = 2743.17, \qquad se(\hat{\beta}_2) = 4086.09$$

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Testing of $\hat{\beta}_0$

CRITICAL VALUE APPROACH

Step -1 : *Formulation of Hypotheses*

$H_0 : \beta_0 = 0$

$H_1 : \beta_0 \neq 0$

Step -2 : *Level of Significance*

$\alpha = 0.05$

Step -3 : *Test Statistic*

$$t = \frac{\hat{\beta}_0 - \beta_0 = 0}{se(\hat{\beta}_0)} = \frac{224928.57 - 0}{5016.15} = 44.84$$

Step -4 : *Critical* Re *gion*

$$|t| \geq t_{\frac{\alpha}{2}(n-p)} \quad \Rightarrow |t| \geq t_{(0.025)(5)} \Rightarrow |t| \geq 2.571$$

Step -5 : *Conclusion*

Since calculated value of t lies in CR so reject $H_{0.} \Rightarrow \beta_0$ *is significant.*

P-VALUE APPROACH

Step -1 : *Formulation of Hypotheses*

$H_0 : \beta_0 = 0$

$H_1 : \beta_0 \neq 0$

Step -2 : *Level of Significance*

$\alpha = 0.05$

Step -3 : *Test Statistic*

$$t = \frac{\hat{\beta}_0 - \beta_0 = 0}{se(\hat{\beta}_0)} = \frac{224928.57 - 0}{5016.15} = 44.84$$

Step -4 : $P - value$

$$p = 2[\Phi(|t_0|)] = 2[\Phi(|44.84|)] = 0.006$$

Step -5 : *Conclusion*

Since $p - value < \alpha$ *so reject* $H_{0.} \Rightarrow \beta_0$ *is significant.*

CONFIDENCE INTERVAL APPROACH

Step -1 : *Formulation of Confidence Interval*

The $(1-\alpha)100\%$ *CI for* μ *is given by*

$$\hat{\beta}_0 - t_{\frac{\alpha}{2}(n-p)}.se(\hat{\beta}_0) \leq \beta_0 \leq \hat{\beta}_0 + t_{\frac{\alpha}{2}(n-p)}.se(\hat{\beta}_0)$$

$$224928.57 - (2.571)(5016.15) \leq \beta_0 \leq 24928.57 - (2.571)(5016.15)$$

$$212030.50 \leq \beta_0 \leq 237826.63$$

Step -2 : *Conclusion*

Since $\beta_0 = 0$ *does not lie in CI so reject* $H_{0.} \Rightarrow \beta_0$ *is significant*

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Testing of $\hat{\beta}_1$

CRITICAL VALUE APPROACH

Step -1: *Formulation of Hypotheses*

$H_0 : \beta_1 = 0$

$H_1 : \beta_1 \neq 0$

Step -2: *Level of Significance*

$\alpha = 0.05$

Step -3: *Test Statistic*

$$t = \frac{\hat{\beta}_1 - \beta_1 = 0}{se(\hat{\beta}_1)} = \frac{15314 - 0}{2743.18} = 5.58$$

Step -4: *Critical* Re *gion*

$$|t| \geq t_{\frac{\alpha}{2}(n-p)} \quad \Rightarrow |t| \geq t_{(0.025)(5)} \Rightarrow |t| \geq 2.571$$

Step -5: *Conclusion*

Since calculated value of t lies in CR so reject $H_0 \Rightarrow \beta_1$ *is significant.*

P-VALUE APPROACH

Step -1: *Formulation of Hypotheses*

$H_0 : \beta_1 = 0$

$H_1 : \beta_1 \neq 0$

Step -2: *Level of Significance*

$\alpha = 0.05$

Step -3: *Test Statistic*

$$t = \frac{\hat{\beta}_1 - \beta_1 = 0}{se(\hat{\beta}_1)} = \frac{15314 - 0}{2743.18} = 5.58$$

Step -4: $P - value$

$$p = 2[\Phi(|t_0|)] = 2[\Phi(|5.58|)] = 0.005$$

Step -5: *Conclusion*

Since $p - value < \alpha$ *so reject* $H_0 \Rightarrow \beta_1$ *is significant.*

CONFIDENCE INTERVAL APPROACH

Step -1: *Formulation of Confidence Interval*

The $(1-\alpha)100\%$ *CI for* μ *is given by*

$$\hat{\beta}_1 - t_{\frac{\alpha}{2}(n-p)}.se(\hat{\beta}_1) \leq \beta_1 \leq \hat{\beta}_1 + t_{\frac{\alpha}{2}(n-p)}.se(\hat{\beta}_1)$$

$$15314 - (2.571)(2743.18) \leq \beta_1 \leq 15314 + (2.571)(2743.18)$$

$$8261.28 \leq \beta_1 \leq 22366.72$$

Step -2: *Conclusion*

Since $\beta_1 = 0$ *does not lie in CI so reject* $H_0 \Rightarrow \beta_1$ *is significant*

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Testing of $\hat{\beta}_2$

CRITICAL VALUE APPROACH

Step -1 : *Formulation of Hypotheses*

$H_0 : \beta_2 = 0$

$H_1 : \beta_2 \neq 0$

Step -2 : *Level of Significance*

$\alpha = 0.05$

Step -3 : *Test Statistic*

$$t = \frac{\hat{\beta}_2 - \beta_2 = 0}{se(\hat{\beta}_2)} = \frac{10957 - 0}{4086.09} = 2.68$$

Step -4 : *Critical* Re *gion*

$$|t| \geq t_{\frac{\alpha}{2}(n-p)} \quad \Rightarrow |t| \geq t_{(0.025)(5)} \Rightarrow |t| \geq 2.571$$

Step -5 : *Conclusion*

Since calculated value of t lies in CR so reject $H_{0.} \Rightarrow \beta_2$ *is significant.*

P-VALUE APPROACH

Step -1 : *Formulation of Hypotheses*

$H_0 : \beta_2 = 0$

$H_1 : \beta_2 \neq 0$

Step -2 : *Level of Significance*

$\alpha = 0.05$

Step -3 : *Test Statistic*

$$t = \frac{\hat{\beta}_2 - \beta_2 = 0}{se(\hat{\beta}_2)} = \frac{10957 - 0}{4086.09} = 2.68$$

Step -4 : $P - value$

$$p = 2[\Phi(|t_0|)] = 2[\Phi(|44.84|)] = 0.021$$

Step -5 : *Conclusion*

Since $p - value < \alpha$ *so reject* $H_{0.}$

CONFIDENCE INTERVAL APPROACH

Step -1 : *Formulation of Confidence Interval*

The $(1-\alpha)100\%$ *CI for* μ *is given by*

$$\hat{\beta}_2 - t_{\frac{\alpha}{2}(n-p)}.se(\hat{\beta}_2) \leq \beta_2 \leq \hat{\beta}_2 + t_{\frac{\alpha}{2}(n-p)}.se(\hat{\beta}_2)$$

$$10957 - (2.571)(4086.09) \leq \beta_2 \leq 10957 + (2.571)(4086.09)$$

$$451.66 \leq \beta_2 \leq 21462.34$$

Step -2 : *Conclusion*

Since $\beta_2 = 0$ *does not lie in CI so reject* $H_{0.} \Rightarrow \beta_2$ *is significant*

HAZBER SAMSON